

MATH0110 Analysis for Joint Honours

<i>Year:</i>	2025–2026
<i>Code:</i>	MATH0110
<i>Level:</i>	4 (UG)
<i>Normal student group(s):</i>	First year students on BSc/MSci Mathematics and Statistical Science only
<i>Value:</i>	15 credits (= 7.5 ECTS credits)
<i>Term:</i>	1
<i>Assessment:</i>	The final weighted mark for the module is given by: 70% Final examination, 15% coursework, 10% mid-session examination and 5% Proofs and Foundation coursework. In order to pass the module you must have at least 40% for both the final examination mark and the overall weighted mark.
<i>Normal Pre-requisites:</i>	A* in A-level Mathematics and Further Mathematics
<i>Lecturer:</i>	Dr M Karpukhin

Course Description and Objectives

This module is an introduction to mathematical analysis, one of the most important and well-developed strands of pure mathematics with many elegant and beautiful theorems. The aim is to introduce students to the ideas of formal definitions and rigorous proofs (one of the fundamental features of modern mathematics, and something that is not familiar from A-level), and to develop their powers of logical thinking. This module leads on to second and third year modules in mathematical analysis and related areas. It is intended for first year students on the Mathematics and Statistical Science degree.

Recommended Texts

Haggarty, Fundamentals of Mathematical Analysis (2nd edition).

Other recommended books are

- (i) Binmore, Introduction to Mathematical Analysis (CUP);
- (ii) M. Spivak, Calculus (Publish or Perish);
- (iii) R. Bartle and D. Sherbert, Introduction to Real Analysis (Wiley);
- (iv) M H Protter and C B Morrey, A first course in real analysis (Springer).

Detailed Syllabus

1. Basic properties of $\mathbb{R}$.
2. Sequences: convergence/limits, limit laws, sandwich theorem, boundedness, monotonicity.
3. Series, “telescopic” series, standard examples, tests of convergence.
4. Limits and continuity: sequential and $\epsilon - \delta$ definition of limit, limit laws, sandwich theorem, limits and composition of functions, Intermediate Value Theorem, Extreme Value Theorem.
5. Differentiability: definition, differentiation rules, examples of non-differentiable functions
6. Mean Value Theorem: Mean Value Theorem, Rolle’s Theorem, and theorems about monotonicity, extrema, and local extrema.
7. Integration: Definitions, elementary properties.

8. The Fundamental Theorem of Calculus.

9. Power series: radius of convergence, Taylor's formula, Taylor's series, power series as functions: differentiability, standard power series (exponential function, $\sin x$, $\cos x$, $\sinh x$, $\cosh x$)